

The Historical Income Panel for the Netherlands

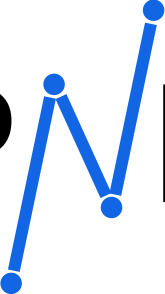
Auke Rijpma

February 2, 2026

- ▶ Economic and social historian at Utrecht University.
- ▶ Interested in reconstructing how past economies and societies worked, and use social scientific theories and methods to do so.
- ▶ In particular interested in the long turn development of welfare & its drivers.
- ▶ Veteran of Clariah, Clariah Plus projects, longstanding interest in developing the infra needed to do the above.
- ▶ PI of the HIPNL project (PDI-SSH).

HIPNL

HIPNL

A stylized graphic of the letters "HIPNL". The letters "H", "I", "P", and "L" are in a bold, black, sans-serif font. The letter "N" is represented by a blue line connecting four blue dots. The dots are positioned at the top-left, top-right, bottom-left, and bottom-right of the "N" shape. The line connects the dots in a sequence that forms the "N" shape: from the bottom-left dot to the top-left dot, then to the top-right dot, then to the bottom-right dot, and finally to the top-right dot again.

Today

- ▶ Look back and forward for the Historical Income Project for the Netherlands, 1850-1920 (HIPNL).
 - ▶ Motivation.
 - ▶ Fit in Dutch historical infra.
 - ▶ Current state of project.
 - ▶ Methods, and changing role of AI.
 - ▶ Future plans.

Motivation

442

W. Salverda and A. B. Atkinson

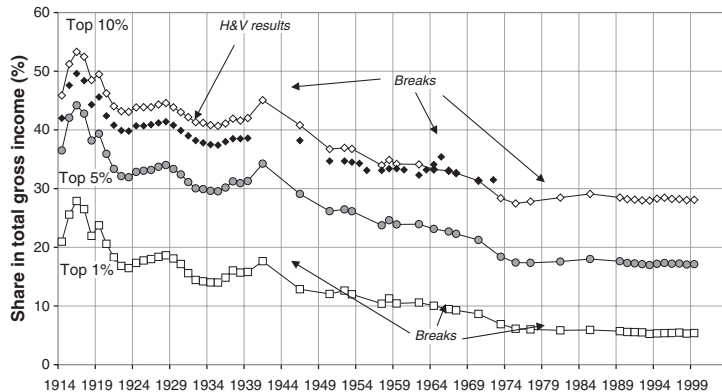


Figure 10.3A Gross income shares of top 10%, 5%, and 1%, Netherlands 1914–99

Figure 1: Income share in Netherlands

Motivation

- ▶ HIPNL strives to address:
 - ▶ Inequality gap: despite renewed interest in inequality, period prior to WWI largely unknown.
 - ▶ Economic history gap: despite importance of microdata in econhis research, we lack *economic* microdata (unlike situation in social and demographic history).
 - ▶ Historical social graph/grail quest: it is possible to reconstruct the demographic, social, and economic lives and connections of all Dutch citizens going back to at least 1812, but we have not done so yet.

Motivation

- ▶ HIPNL contributes to this by:
 1. collecting 100.000s of household-level income estimates for the period 1850–1920 on the basis of the *Hoofdelijke Omslag* taxes.
 2. Linking these households to the Dutch population registers (*Bevolkingsregisters*), in turn allowing link to civil registry as a central hub in the Dutch historical microdata infrastructure.

HIP-NL sources: Zaandijk 1879

[illegible]

HIP-NL sources: Utrecht 1919

Volg. nummer.	Naam, beroep en woonplaats der belastingsschuldigen	Burg. staat en aantal kinderen.	Werkelijk inkomen.	Aftrek levensonderhoud.	Belastbaar inkomen.	AANSLAG.	
						Aantal maanden.	Bedrag.
3511	Mej. G. v. Leeuwen Vert.: M. v. Leeuwen, forstelijk Agnesthuys, N° 2	0/-	950	500	450		25 22
2	Wed. H. M. Weijzenberg geb. W/- T. Gardet Amsterdam, Florastr. 29. N° 2	1750	550	1200			77 41
3	G. F. W. Bantz Cur: G. Bantz Goheveningen, Haringkade 113. N° 2 <i>Geo: naam Antwerpen</i>	0/-	800	500	300		15 25
4	C. Ripping Cur: Jhr. C. T. Goldman te Eindhoven N° 2 <i>Antlaag Arnhemmed. N° 80</i>	0/-	1150	500	650		38 09
5	K. M. Bikkema Veider Vert: G/- J. J. Wieringa, huisgezand N° 2	0/-	950	500	450		25 22
6	D. J. A. Tol Cur: Mevr. D. J. A. A. Tol geb. A. Zimmerman Franeker, Zuideweg 90. N° 2	14/3	2100	700	1400		92 23
7	J. v. d. Wijk Prov. bevelind, G/- H. G. v. d. Wiik te Hilversum N° 2	0/-	2950	500	2450		172 96
8	J. J. Wilmenga Vert: J. J. A. Wilmenga te Garijp N° 2	0/-	900	500	400		22 16
9	A. Erainsheerd Klerk S. S. Amsterdamstr. N° 4bis	0/-	650	500	150		2 66
3520	Mej. E. A. Huijgens Handw. Oudewijzeres	0/-	700	500	200		10 59

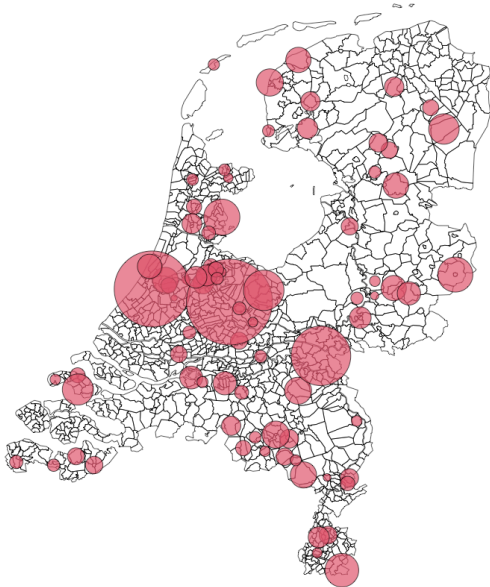
HIP-NL sources: Leiden 1889

	7	Klasse.	Inkomen.	1889.	1888.	Belasting.
111 J. P. A. Mulert	8	1375	25,35			
113 G. J. Bos		4000	93,60	87,10		
115 J. Brevée	13	3250	74,10	74,10		
115 T. van Veen	3	550	3,90	3,90		
117 J. van der Staaij	1	425	0,65	0,65		
Nonnensteeg.						
1 W. F. R. Suringar	23	15000	372,—	372,—		
5 F. J. Struikenkamp	2	475	1,79	1,79		
7 H. Witte	12	2750	61,10	61,10		
7 H. W. Witte	5	775	9,75	10,40		
7 E. T. Witte	3	550	3,90	2,60		
2 A. Teske	1	425	0,65	0,65		
16 P. Brieër	1	425	0,65	0,65		
18 F. C. J. Schnabel	1	425	0,65	0,65		
18 J. T. Schnabel	1	425	0,65	0,65		
20 J. Evers		600	4,88	4,88		
Kolfmakerssteeg.						
4 A. van der Vecht	4	650	6,50	6,50		
7 A. Versnij	2	475	1,87	1,84		
14 W. van Ingen	1	425	0,59	0,59		
24 C. T. H. Mook	3	550	3,58	0,62		
Zegersteeg.						
1 G. van der Reijden	1	425	0,65	0,65		
Doelenachtergracht.						
2 H. Frankhuijzen	1	425	0,65	0,65		
19 J. Ober	1	425	0,62	0,62		
Kaiserstraat.						
1 J. van der Poel	6	925	12,29	12,29		
1 P. B. van Oerle	3	550	3,90	3,90		
3 J. F. Neuteboom	1	425	0,65			
3 C. H. Beij	2	475	1,95			
5 W. F. K. Buitenweg	1	425	0,65	0,65		
5 J. van Ulden	1	425	0,65	0,62		
7 W. Epskamp	5	775	8,58	9,75		
7 J. N. van der Mark		900	12,22	12,22		
9 H. L. van Nifterik	12	2750	59,87	61,10		
11 Wed. J. v. d. Meulen-Wakerlin	1	425	0,62	0,62		
13 D. Jurriaanse	8	1375	25,35	25,35		
13 J. Groot	7	1125	18,85	18,85		
19 C. Collee	3	550	3,58	3,66		
27 D. Segeyn	1	425	0,62	0,62		
33 L. de Leeuw	1	425	0,62	0,62		
39 T. N. Schotman	3	550	3,90			
47 H. H. van Barmburgh		4000	44,97	48,09		

In the larger infrastructure.

- ▶ HSN: the original large historical microdata project. HIPNL adds income estimates to 10% of HSN's observations.
- ▶ LINKS: all births, marriages, and deaths. All linked observations in HIPNL will be connected to LINKS
- ▶ CLARIAH: Linked Data Workbench for SSH + LD instance of LINKS, in which HIPNL can be integrated.
- ▶ Lifetree-NL: ambition to add HSN, LINKS, HIPNL, etc. to present-day SSD.
- ▶ Archival scans and indexing projects: HIPNL would be impossible without investments by archives to digitise and index their materials.

HIP-NL sample



Current state

- ▶ Sample of municipalities, stratified by province, municipality size, covering 10% of population in 7 sample years.
- ▶ Resulting database has income and tax data for 350k households, from 78 municipalities.
- ▶ Linked to population and civil registries for c. 50% of data.

Methods

- ▶ At first, classic #econhis project:
 - ▶ Lots of RAs scanning and transcribing scans.
 - ▶ Traditional OCR for the printed volumes.
 - ▶ ML (boosting) for the linkage.
- ▶ HIPNL coincided with the LLM revolution, and we adapted our approach to the new possibilities.

Methods

- ▶ OCR: traditional issues: character error rate good enough, but post-processing very time intensive.
- ▶ Worked on NOCR, direct information extraction from scans using VLMs.
 - ▶ DONUT: finetune our own pre-VLM to directly create structured data (promising but still time-intensive).
 - ▶ In the end, leverage commercial VLMs (Gemini) in a few-shot learning approach.

187						W I J K					
Volgnummer.	N A M E N VAN DE BELASTINGSSCHULDIGEN.	WOONPLAATS. Straat, gracht, enz. en nummer.	Klasse.	Bedrag van den aanslag à 3 pet.		N ^o . van het					
11061.	G. Fukkink	Oudekamp N ^o . 10	2	1	4 50	111	14	59			
11062.	C.B. Beijerman	N ^o . 12		10	60	53					
11063.	J. van Hijtveld	N ^o . 16	2		4 50	15					
11064.	J.C.E. Procksie	N ^o . 16	1		1 50	74					
11065.	G. van Baak	N ^o . 10	9		49 50	8					
11066.	J.H. Kneebout	Oudekerkhof N ^o . 3	1		1 50	45					
11067.	J. van der Pijl		2		4 50	112					

{

```

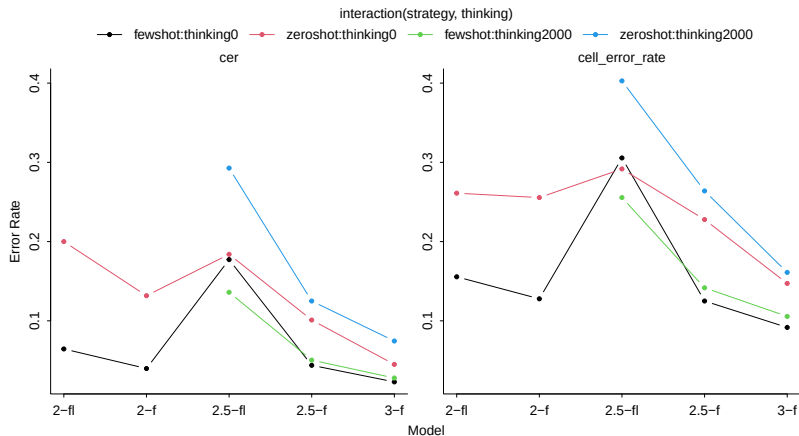
"volgnummer": 1861,
"title": null,
"initials": "G.",
"surname": "Fukkink",
"maiden_name": null,
"street": "Oudekamp",
"house_number": "10",
"class": 2,
"tax": 4.5

```

}

NOCR: error rates

Error Rates by Model, Strategy, and Thinking Budget



NOCR

- ▶ Able to circumvent nearly all postprocessing, left with only more straightforward character corrections.
- ▶ Massive increase in productivity without loss in quality.
- ▶ Low cost (back then: free; now: 1000s of scans for c. \$1)
- ▶ Cornerstones:
 - ▶ Strong image comprehension of Gemini models
 - ▶ Strong schema adherence and prompt mapping.
 - ▶ In-context consistency: identifier perfectly remembered (even 2bdf4d0c-7ce1-40f6-aceb-0b85a2768881)

Further LLM use

- ▶ Categorise (open weights, facebook/huggingface) names as male/female.
- ▶ Link checks:
 - ▶ Built an ML-based (gradient boosting) approach
 - ▶ But verified links with LLMs combined with human review.
 - ▶ Good at dealing with missing data, determining ambiguous link candidates.
 - ▶ Probably superior if expensive method.

AI reflections

- ▶ While using AI tools boosts productivity, there are drawbacks.
 - ▶ Commercial closed v. commercial open v. open source
 - ▶ Reproducibility
 - ▶ Costs
 - ▶ Privacy
- ▶ Good scans, open data (potential training/finetuning) and compute are key to enable these pipelines.

Future plans

- ▶ HIPNL ambitions:
 - ▶ Ingesting newly available data such as population registers, tax records that keep coming online.
 - ▶ Finish connections to other historical microdata to make HIPNL a broader “welfare” panel (health, wealth, education).